Time-of-Arrival Measurement Using Adaptive CMOS IR-UWB Range Finder With Scalable Resolution

Tae Hwan Jin, *Student Member, IEEE*, Hong Gul Han, *Student Member, IEEE*, and Tae Wook Kim, *Senior Member, IEEE*

***Abstract*—This paper introduces an adaptive CMOS impulse-radio ultra-wideband (IR-UWB) range finder with scalable resolution for localization applications and presents a brief demonstration of its operation. The distance resolution of the proposed range finder can be adapted to the ranging environment using a two-step time-to-digital converter (TDC) with scalable delay cells. Using the scalable time resolution of the TDC, the proposed range finder can control its maximum detectable distance and range resolution without any change in hardware. The proposed CMOS IR-UWB range finder has been implemented with a 0.11-μm process; the effective areas of the transmitter and receiver are 0.065 mm^2 and 0.27 mm^2, respectively. The range finder has a range 8 m, with 2.9 mm scalable resolution. The power consumptions of the transmitter and receiver are 1.5 mW and 42 mW, respectively.**

***Index Terms*—Impulse radio, ranging, scalable resolution, time-of-arrival, time-to-digital converter, ultra-wide band.**

I. Introduction

SMART mobile devices with their fast developing wireless communication technologies are becoming increasingly popular, as they allow users not only to enjoy faster data communications, but also to expect the possibility of using their multiple functions in various other applications. One of these applications is ranging. In particular, location-based applications such as indoor GPS, 3D virtual games, wireless rulers, pre-collision avoidance systems, and warehouse automation are attracting much interest from the industry.

There have been many approaches to wireless ranging, as listed in Table I. Frequency modulated continuous wave (FMCW) systems can be used to detect objects at long distances but are not appropriate for measuring distances with high accuracy, especially in multipath environment [1]–[3]. Ultra-wideband (UWB) is the most promising standard for wireless ranging, mainly because of the innate characteristics of its signal, which uses a ultra-wide frequency domain spectrum, allowing for high integrity and performance in multipath conditions. Previous studies [4]–[11] have demonstrated high-accuracy ranging UWB systems with excellent performance even in a multipath environments.

TABLE I
Comparison of Previous Research and Commercial Ranging Systems

Ref	System	Inaccuracy /Effective coverage from the evaluation	Freq. [Hz]	Implementation	Dim
[1]	FMCW	0.1 mm / 5 mm	35 G	MMIC	1D
[2]	FMCW	15 mm * / 36 m	5.8 G	SiP	1D
[3]	FMCW	50 mm / 5.5 m	24 G	0.13-μm CMOS	1D
[5]	UWB	0.69 mm *, 1.49 mm ** / 40 cm	5 G-11 G	System (MIC +Commercial MMIC +Module +FPGA)	1D
		0.69 mm *, 2.61 mm ** / 40 x 40 cm^2			2D
		0.93 mm *, 2.45 mm ** / 40 x 40 x 40cm^3			3D
[6]	UWB	2.66 mm * / 40 cm	5 G-11 G		1D
		1.98 mm*** / 1.4 m x 1.9 m x 2.5 m			3D
[7]	UWB	1.7 cm * / 10 m	7 G-8 G	System	1D
[8]	UWB	0.6 cm *, 0.5 cm ** / 7.4 m	3 G-5 G	System	1D
[9]	UWB	25 mm / 1 m	250 M-500 M	0.18-μm CMOS	1D
[10]	UWB	14 mm / 10 m	0-960 M	0.13-μm CMOS	1D
[11]	UWB	9.7 mm / 1 m	3 G-4 G	0.13-μm CMOS	1D

*: Standard deviation
**: Mean error
***: RMSE referred to Optotrak 3020
MIC : Microwave integrated circuit

One of the remaining challenges in designing a UWB ranging system is the issue of channel modeling in such multipath environments [4]–[6]. Therefore, leading-edge detection method has been employed in UWB ranging systems to improve their immunity to multipath effects [12]–[15]. The architecture used in [11] adopts leading-edge detection with a time-to-digital converter (TDC). In this architecture, the first received impulse which indicates the signal arriving through the direct path is converted into a digital pulse. The TDC then measures the time gap between a reference clock and the converted pulses with a certain time resolution, and outputs digital codes corresponding to the measured time gaps, making the system architecture and its operation very simple and intuitive.

However, the ranging system in [11] uses a fixed resolutions for distance measurement, which implies that the maximum detectable range is fixed as well. As a result, additional or

Manuscript received April 15, 2016; revised June 10, 2016; accepted June 20, 2016. Date of publication September 19, 2016; date of current version September 27, 2016. This work was supported by the Space Core Technology Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Science, ICT and Future Planning (NRF-2014M1A3A3A02034806), and by the IDEC. This paper was recommended by Associate Editor T. Serrano-Gotarredona.

The authors are with School of Electrical and Electronic Engineering, Yonsei University, Seoul 120-749, South Korea (e-mail: taewook.kim@yonsei.ac.kr).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TCSI.2016.2583661

1549-8328 © 2016 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

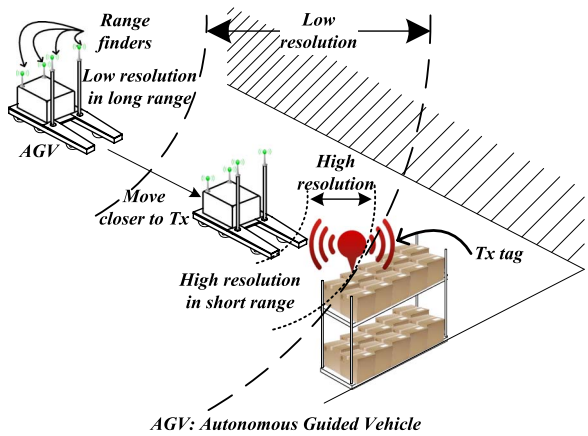


Fig. 1. AGV for a warehouse automation application for the proposed adaptive IR-UWB ranging system with scalable resolution.

extended circuitry is needed to extend its dynamic range, increasing system complexity. If we employ the approach of human recognition of distance to the wireless ranging system in [11], we can extend both the detection range and ranging resolution without major difficulties. In many cases, a more precise distance resolution is usually required to measure distances to close objects, but the resolution requirements can be alleviated when measuring longer distances. If this concept is adopted for the impulse-radio UWB (IR-UWB) range finder, the ranging techniques can become more efficient in many applications. The autonomous guided vehicle for warehouse automation shown in Fig. 1 is an example of such an application, in which the IR-UWB range finder receiver finds the target by changing the resolution of distance measurement continuously.

If the target is far away, the range finder covers a longer range with coarse resolution; if the target is close, the range finder covers only a short range, but with fine resolution. This can be achieved using a scalable time resolution in the TDC in [11]. With such a scalable time resolution, the TDC can adjust the maximum detectable time range without modifying its dynamic range. This capability in the time domain directly maps to an equivalent capability in the distance domain as well.

In this study, a TDC-based wireless range finder with extended measurement rage and scalable resolution is proposed, using a two-step TDC with scalable time resolution. The range finder is capable of high resolution for short distances and can also measure long distances, albeit with a coarser resolution. To compensate for error curvatures in the time-of-arrival (ToA) measurements of the range finder, a digital curvature correction has been performed.

This paper is organized as follows. Section II briefly demonstrate the concept of ranging operations with fixed and scalable resolutions. Section III discusses the architecture of the proposed adaptive IR-UWB range finder and its operation. Section IV discusses the distance error analysis of the IR-UWB range finder. Section V discusses system implementation and presents the measurement results for the TDC, and ToA. Chip-to-chip variations are presented, and a calibration technique is discussed. Finally, Section VI summarizes the potentialities of this study and concludes the paper.

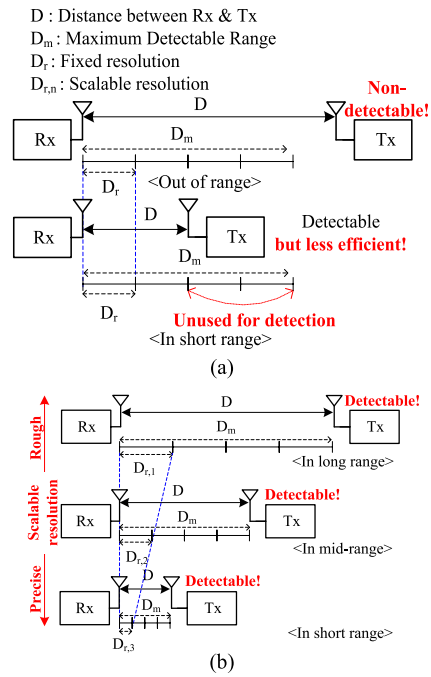


Fig. 2. Range detection techniques with (a) conventional fixed resolution and (b) proposed scalable resolution.

II. Range Operation With Scalable Resolution

In the IR-UWB range finder based on the TDC, the maxi mum detectable distance D_m can be calculated as follows:

$$D_m = D_r \times 2^n \tag{1}$$

where D_r is the distance resolution of the TDC, n is TDC bits. Only two factors are needed to increase the maximum detectable distance: the distance resolution and the dynamic range (2^n) of TDC. Increasing the dynamic range with a fixed resolution leads to higher area cost and additional complexities to the circuit, such as a longer delay line. The maximum detectable distance, however, is improved without increasing the dynamic range as long as the distance resolution is increased. Because the distance to a target located in a farther range is measured with larger distance resolution, the accuracy is degraded. In common applications, however, the farther away the target is, the less significant the resolution or, in turn, accuracy becomes.

The equation and concept for the control of distance resolution are shown in Fig. 2, where a conventional operation with a fixed distance resolution (D_r) and the proposed operation with a scalable distance resolution $(D_{r,n})$ are depicted. The following two cases are good examples for the comparison between the two types: the transmitter located out of the maximum detectable distance and the transmitter located in short range. In the former case with the conventional operation depicted in (a), the receiver of range finder cannot measure the distance (D) to the transmitter. In the latter case, the receiver can sufficiently measure the distance, but the occurrence of the redundant codes results in lower efficiency in hardware perspective. Redundant codes inevitably requires redundant delay line which occupies

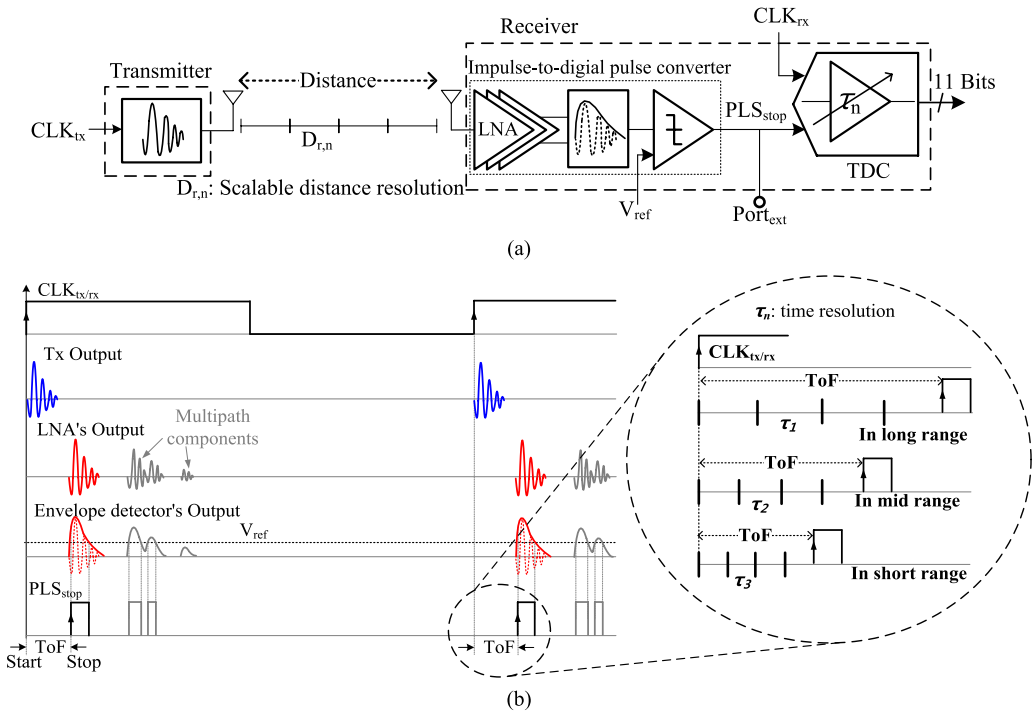


Fig. 3. (a) Architecture for the adaptive IR-UWB range finder, and (b) its timing diagram.

large silicon area. On the other hand, in the operation with a scalable resolution shown in (b), the receiver can measure the distance in both cases, even without any redundant codes. The receiver chooses medium resolution $(D_{r,2})$ to detect the transmitter in mid-distance, rougher resolution $(D_{r,1})$ to detect one at a longer distance, and more precise resolution $(D_{r,3})$ to detect one at a shorter distance.

Finally, even though there are two systems that have the same dynamic range of 2 bits, the ranging system with a scalable resolution usefully operates with higher efficiency and allows flexibility in its operation to be considerably free from the limit of the maximum detectable range.

III. Architecture of Adaptive IR-UWB Range Finder

Fig. 3 shows (a) a diagram of the proposed adaptive IR-UWB CMOS range finder, and (b) a timing diagram of the operation. The transmitter is designed as a digital-impulse-generator type in [16] and [17] and generates an impulse signal at the rising edge of the input clock of the transmitter (CLK_{tx}). The receiver of the adaptive IR-UWB range finder is designed and operated mostly according to that in [11]. The low-noise amplifier (LNA) and a four-stage preamplifier receives and amplifies the impulse signal transmitted from the transmitter. The envelope of the amplified signals is detected by the envelope detector and then converted into a digital pulse if the voltage of the envelope is higher than V_{ref}, the reference voltage of the comparator. Finally, the TDC digitizes the time interval between the rising edges of $\mathrm{CLK}_{tx/rx}$ and $\mathrm{PLS}_{\mathrm{stop}}$, which is time-of-flight (ToF), with a scalable time resolution (τ_n) corresponding to the scalable distance resolution $(D_{r,n})$ in Fig. 2. For example, the TDC digitizes a long time-of-flight with a long time resolution (τ_1) and a short time-of-flight with a short time resolution (τ_3).

The determining factor of the time resolution of the TDC is the delay time of the unit delay cell. A conceptual diagram of the TDC using a basic delay line, its actual coarse unit delay (τ_c) cell used in this work, and its delay simulation are shown in Fig. 4. Two identical delay buffers compose one actual coarse unit delay cell, and an arbiter senses the output of the last delay buffer in the coarse unit delay cell. Each delay buffer in the coarse unit delay cell has half $(0.5\tau_c)$ of the entire coarse unit delay. The delay of the buffers depends on the bias current, which is controlled by the bias current of the global current mirror. Thus, we can finally scale the distance resolution of the range finder using the TDC with scalable time resolution.

A coarse-fine TDC with scalable resolution is presented in Fig. 5. The design of the coarse-fine TDC is based on two-step architecture [18], [19]. Prior the core operation of the TDC, the duty cycle of the input clock (CLK_{rx}) of the receiver is expanded by the duty expander (DE) to the rest of the cycle, except for approximately 2 ns of the so-called "dead zone" at the beginning. The DE is simply composed of an inverter and a NAND gate in this design, and the TDC does not operate during the dead zone. T flip-flop is used between the comparator and the TDC to keep $\mathrm{PLS}_{\mathrm{stop}}$ "high" until the next $\mathrm{PLS}_{\mathrm{stop}}$ is generated at the falling edge of CLK_{tx} in [11], but not in the proposed range finder. Logically, a malfunction can occur in the TDC because of the operation of the T flip-flop when a multipath component is detected after the direct path signal by the comparator in [11]. The perfect solution to neglect the multipath components is to keep $\mathrm{PLS}_{\mathrm{stop}}$ "high" from the first rising edge that indicates the direct path signal until the end

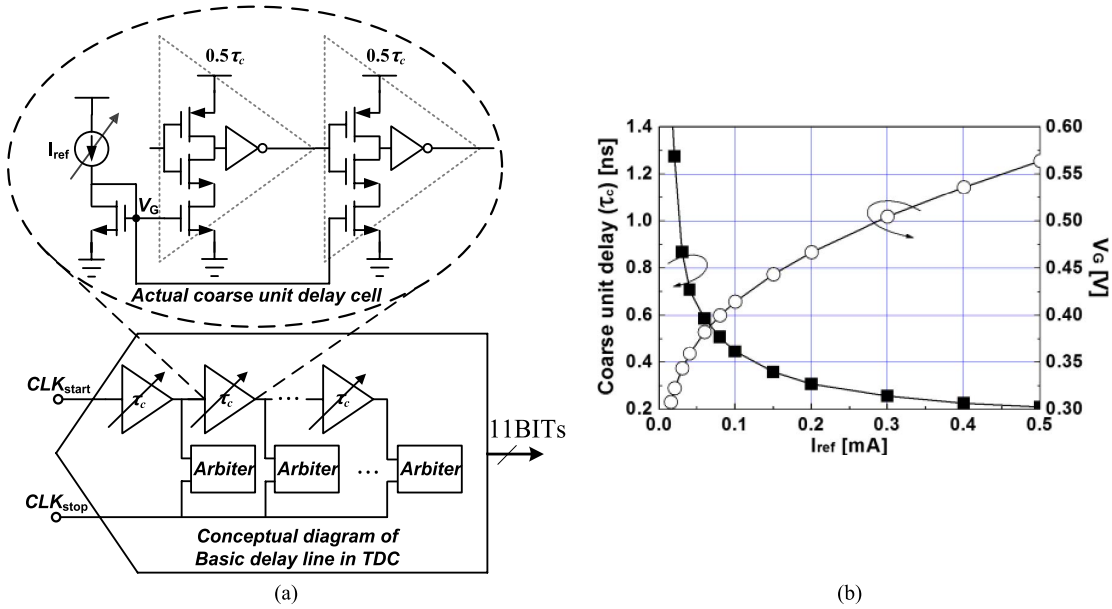


Fig. 4. (a) Schematic of coarse unit delay cell of a TDC with scalable resolution. (b) Simulation results of delay (τ_c) and bias voltage (V_G), depending on bias current.

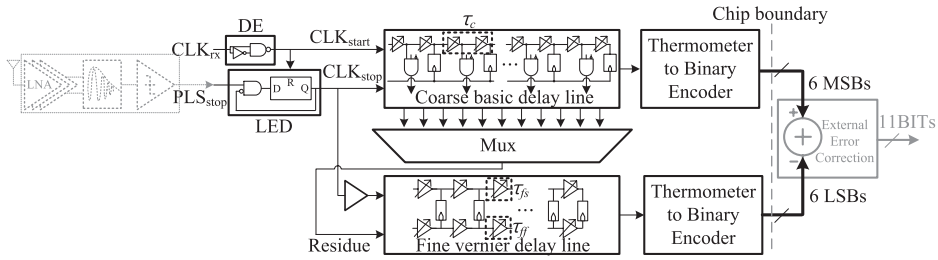


Fig. 5. Block diagram of a two-step TDC.

of a cycle of CLK_{rx}, not until the next $\mathrm{PLS}_{\mathrm{stop}}$. In this study, another type of flip-flop with other gates is used as a leading-edge detector (LED) after the comparator. An inverter, an AND gate, and a D flip-flop combine to form the leading edge detector, which operates only depending on the direct path signal, even in the case of detection of multipath components at the comparator. The D flip-flop outputs "high" at the first rising edge of $\mathrm{PLS}_{\mathrm{stop}}$, and then keeps the node "high" until the falling edge of $\mathrm{CLK}_{\mathrm{start}}$, which resets the D flip-flop.

The core of the two-step TDC is based on the architecture in [18], where a 2D vernier plane [20] is used for fine detection. However, here a conventional vernier delay line [24] is used for fine detection instead, in order to scale the resolution handily. The actual circuit for the fast fine unit delay (τ_{ff}) and the slow fine unit delay (τ_{fs}) in the vernier delay line is identical with one delay buffer ($0.5\tau_c$) in the coarse unit delay cell.

Fig. 6 shows the ranging algorithm for distance measurement with the scalable resolution. The range finder determines ranging resolution with the value of MSB in the TDC. The algorithm can be applied to the range finder with a simple digital logic but in this work it was controlled manually for test purpose.

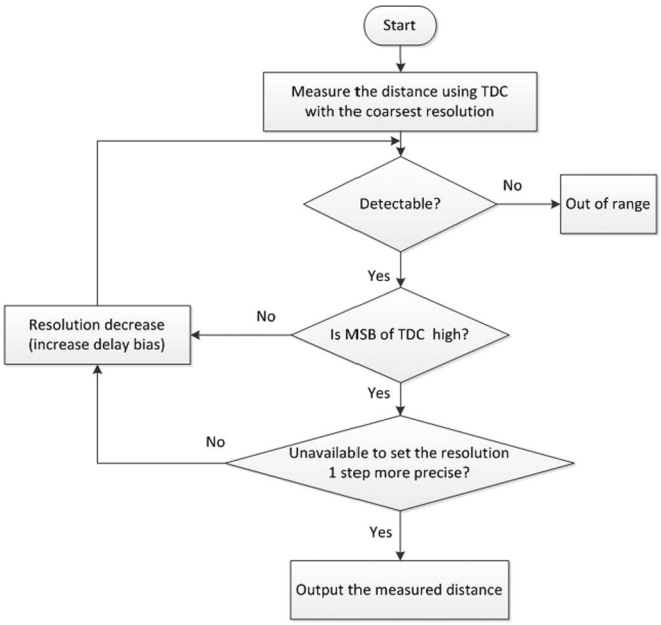


Fig. 6. Ranging algorithm for TDC with scalable resolution.

IV. The Distance Error Analysis

Distance errors in range finder originate from many different sources, such as jitter and clock mismatches, receiver thermal noise, noise and nonlinearities in the TDC, and walk error. The clock error can be ignored in our measurement because both Rx and Tx share the same clock [14]. Even though sharing the same clock may not be a realistic situation in all environment, it is enough to evaluate the proposed range finder because this work can be used as a time difference of arrival (TDoA) estimation algorithm, and hence use a shared clock. Therefore, the considered main error sources are the TDC, thermal noise and walk error.

A. The Distance Error From the TDC

1) TDC INL Error: The TDC errors resulting from the mismatch among delay cells and the delay line jitter can be expressed via the concept of integrated nonlinearity (INL) [20]. The distance error resulting from INL $(\Delta D_{\mathrm{TDC_INL}})$ can be obtained with the following equation:

$$\Delta D_{\mathrm{TDC_INL}} = \tau_r \times 2^{n_{INL}} \times c \tag{2}$$

$$n_{\mathrm{INL}} = \frac{1}{2} \cdot \log_2 \left(\sum_{i=1}^{N_{\mathrm{tot}}} \left(1+\sigma_\alpha^2\right)\left(1+\sigma_\beta^2\right) \right) \tag{3}$$

where τ_r is the time resolution of a the TDC, c is the speed of light. n_{INL} is the bit of INL, N_{tot} is the total number of delay stage, σ_α^2 is the absolute mismatch variance, and σ_β^2 is the local mismatch variance. The INL of a TDC is originated from delay line uncertainties such as mismatches and can therefore be calculated from an absolute mismatch and a local mismatch. Each value of σ_α^2 and σ_β^2 is extracted from Monte Carlo simulation. We can estimates the INL of the TDC using (3). $\Delta D_{\mathrm{TDC_INL}}$ increase with τ_r and INL. In general, INL depends on the TDC architecture; consequently, the scalable TDC has an almost similar INL for all resolutions.

2) TDC Jitter Error: The distance error resulting from the TDC jitter can be expressed by the following equation:

$$\Delta D_{\mathrm{TDC_jitter}} = \left(\sqrt{N} \cdot \sigma_{\mathrm{TDC_jitter}} + \sigma_{Clk_\mathrm{jitter}}\right) \times c \tag{4}$$

$\Delta D_{\mathrm{TDC_jitter}}$ is the distance error resulting from the TDC jitter, $\sigma_{\mathrm{TDC_jitter}}$ is the standard deviation of the jitter noise, $\sigma_{Clk_\mathrm{jitter}}$ is the standard deviation of reference clock jitter, and N is the number of delay stages. $\sigma_{Clk_\mathrm{jitter}}$ is about 11 ps in our measurement case. $\sigma_{\mathrm{TDC_jitter}}$ can be extracted by simulation. $\Delta D_{\mathrm{TDC_jitter}}$ is a result of the thermal noise of the delay line; therefore, the longer delay line is, the more jitter it generates. The proposed system uses the same length of delay line for all ranging modes; the contribution of $\Delta D_{\mathrm{TDC_jitter}}$ to the distance error is hence the same for all ranging modes.

B. The Distance Error From Receiver Thermal Noise

The thermal noise in the receiver, expressed by the signal to noise ratio (SNR), also contributes to the distance error [22]. The received signal is used as one of the inputs to the TDC; the noise from the receiver can therefore be interpreted as equivalent jitter to TDC jitter. If we assume that the received signal level is the same for all distances, then we can translate the receiver thermal noise into a distance error from the TDC with following equation.

The input noise of Rx is given by following equation where k_B is Boltzmann constant, T is absolute temperature and B_{ev} is the bandwidth of the Rx

$$N_{\mathrm{in_Rx}} = k_B T B_{ev}. \tag{5}$$

The output noise of Rx is given by following equation where G is the Gain of Rx, F is the noise factor of Rx

$$N_{\mathrm{out}} = F G k_B T B_{ev}. \tag{6}$$

If we assume the thermal noise fluctuates the rising slope of the Rx output and it introduce changes in first crossing points, the jitter can be approximately represented by following equation [20] where the rising slew rate is assumed $V_p \times B_{ev}$ $[V/s]$:

$$\text{Jitter} = \frac{\text{noise voltage}}{\text{slew rate}} = \frac{\sqrt{F G k_B T B_{ev}}}{V_p \times B_{ev}} \tag{7}$$

$$\Delta D_{\mathrm{RX_thermal}} = \frac{\sqrt{F G k_B T B_{ev}}}{V_p \times B_{ev}} \times c \tag{8}$$

$\Delta D_{\mathrm{RX_thermal}}$ is the timing error resulting from the receiver thermal noise. The jitter variance which will result in a distance measurement error, as discussed above, is obtained by integrating the thermal noise of the receiver to the bandwidth of the envelop detector. The maximum jitter from above calculation is 10.2 ps.

C. Walk Error

The impulse signal reaching the range finder receiver is converted into a positive only signal with an envelope detector. The envelop signal is then converted into a digital pulse by comparator, using a fixed threshold voltage (V_{ref}). Since the comparator generates the digital pulse according to the V_{ref}, not the envelope's output voltage, if the signal envelope has lower voltage, the generated digital pulse will be delayed [23], [25]. This is referred to as walk error, and is a common source of errors in distance measurement systems based the received signal envelope. Fig. 7(a) is a schematic representation of a triangular wave generated by the envelope detector used to calculate the walk error [23]. In this figure, V_p is the peak voltage of the envelope signal and t_p is cross point of the envelope signal and V_{ref}. Pulse_1 and Pulse_2 represent the signals returning from near and far points, respectively. The signal attenuation level of both pulses is naturally determined by the distance, and the positions of the generated digital pulses differ accordingly. As shown in the figure, this enlarges the original (distance related) time difference between the two digital pulses by Δt_p. If we let

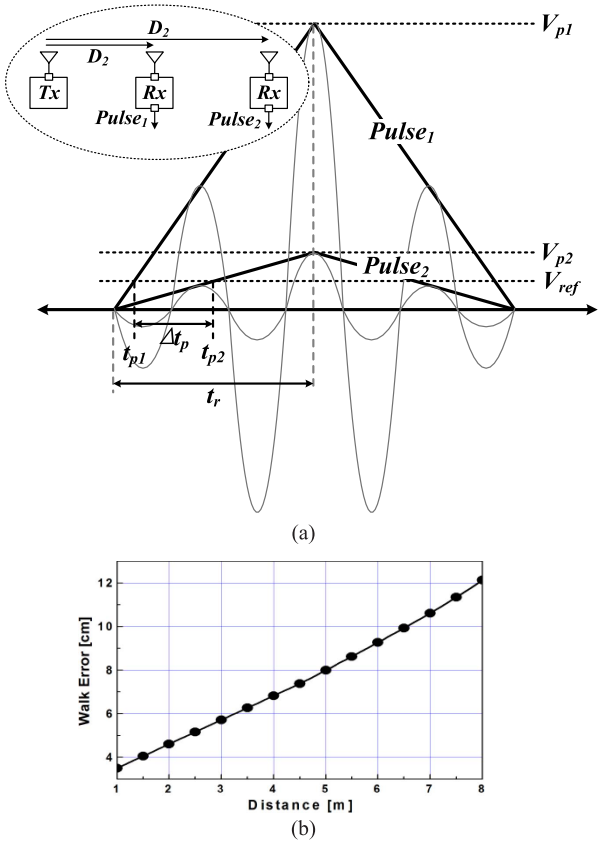


Fig. 7. (a) Concept diagram of the walk error and (b) simulation result.

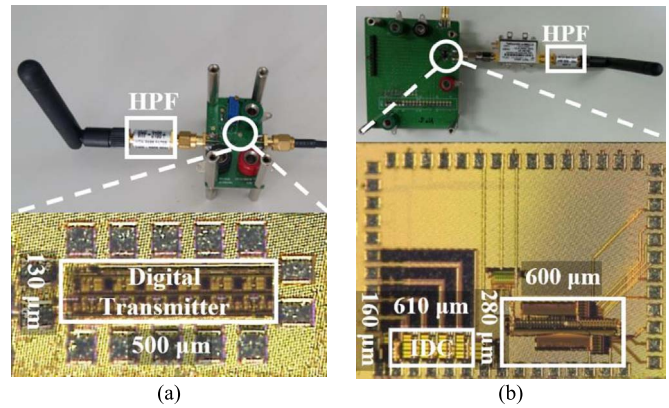


Fig. 8. Chip micrographs of (a) the Tx and its evaluation board, (b) Rx and its evaluation board, and the adaptive IR-UWB range finder.

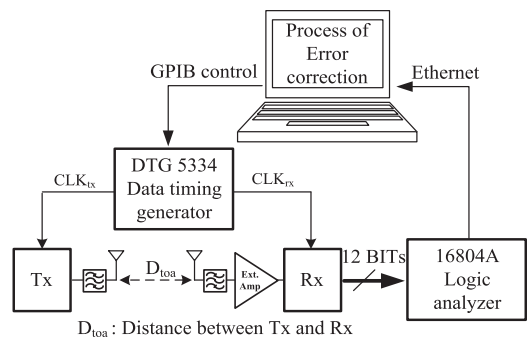


Fig. 9. Test setup for ToA measurement.

the rising time be t_r as in Fig. 7, then the walk error Δt_p can be obtained as follows:

$$\Delta t_p = \frac{V_{\text{ref}}}{V_{p_\max}} \times t_r \times (2\sqrt{\pi} \times D - 1) \tag{9}$$

$$\Delta D_{\text{walk}} = (\Delta t_p + \Delta t_{cst}) \times c \tag{10}$$

$V_{p_\max}$ is the maximum peak voltage of the signal envelope. ΔD_{walk} is the distance error resulting from the walk error. D is the distance between Rx and Tx. Δt_{cst} is a settling time of the comparator over the amplitude of the input signal. Δt_{cst} increases as distance increases because the settling time of a comparator increase as input signal decreases. Fig. 7(b) shows a simulation result of the walk error vs. distance, as obtained from (9) and (10). The walk error increases as distance increases because the signal attenuation increases with distance. For example, if the distance is 5 m the walk error is 8 cm.

Considering the above-mentioned errors, the distance estimated from the range finder can be expressed as

$$D_{\text{cal.}} = D_{\text{real}} + \Delta D_{\text{walk}} + \Delta D_{\text{RX_thermal}} + \Delta D_{\text{TDC_INL}} + \Delta_{\text{TDC_jitter}} \tag{11}$$

where, $D_{\text{cal.}}$ is the calculated distance by the range finder, D_{real} is the real distance. The calculated distance is the sum of the real distance and all distance errors involved in the range finder measurement. The walk error and error from the INL of TDC is deterministic over the distance thus it can be calibrated in post processing. These errors can be properly calibrated, as will be discussed in Section V. However, the distance errors resulting from the receiver thermal noise and the TDC jitter can not be calibrated and will remain as the main contributors to the distance error, even after calibration. However, the error from the thermal noise and TDC jitter is random noise thus those can be reduced by taking average of multiple pulses.

V. Implementation and Measurement Results

The proposed adaptive CMOS IR-UWB range finder was implemented in a 1P6M, 0.11-μm CMOS process. The chip micrographs of the transmitter and receiver for the adaptive IR-UWB range finder are shown in Fig. 8(a) and (b), respectively. Omnidirectional monopole antennas (0 dBi) and external commercial filters (Mini Circuits VHF-3100+) are used in the transmitter and the receiver in order to suppress interferences such as CDMA/Bluetooth/WiFi, and an external commercial amplifier (Mini Circuits ZX60-3800LN-S+, with 25 dB of maximum gain) is used in the receiver for ToA measurement. The transmitter occupies 500 μm $\times$ 130 μm of the core size, which occupies an area of 0.065 mm^2. The receiver occupies 610 μm $\times$ 160 μm of the core size of the impulse-to-digital pulse converter (IDC), and 600 μm $\times$ 280 μm of the core size of the two-step TDC, which occupies a total area of 0.27 mm^2. The setup environment for a ToA measurement of the adaptive CMOS IR-UWB range finder is shown in Fig. 9. Both the

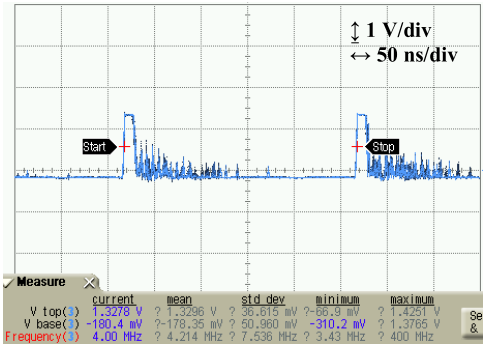


Fig. 10. Comparator output pulse at distance of 2 m between the antennas of the transmitter and receiver.

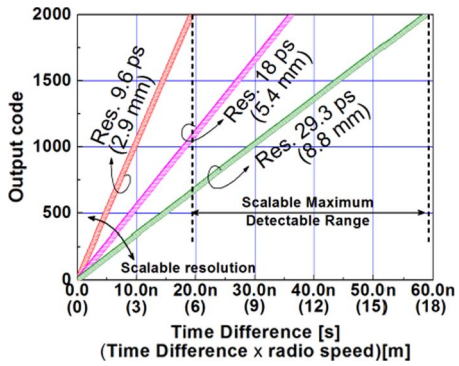


Fig. 11. TDC transfer curves with various time resolutions.

transmitter and the TDC in the receiver are technically synchronized with the same base clock from DTG5334 for the ToA measurement. The data acquisition of digital 12 bits out of the TDC in the receiver is performed by a logic analyzer (Agilent 16804A), and then the acquired 12 bits of data are transferred to a PC. The data then passes through the process of error correction, finally being converted into digital 11 bits. In the following subsections, we show the results of the conversion of the impulse signal into the digital pulse, the operation results of the TDC (which carries out the core principle of the proposed adaptive IR-UWB range finder), the results of the comparison of the ToA measurement and the TDC measurement, and then the chip-to-chip variation with the digital correction technique.

A. Impulse-to-Digital Pulse Converter

Fig. 10 shows the digital pulse converted from the received impulse signals by the impulse-to-digital pulse converter. The pulse was monitored through Port_{ext} when the receiver was 2 m away from the transmitter. The direct path signal decreases as the distance between the receiver and the transmitter gets longer, which causes the digital pulse to become more corrupt and weak.

B. TDC-Only Measurement

Only the operation of the TDC can be evaluated by connecting the output of a data-timing generator (Tektronix DTG5334) to Port_{ext} in Fig. 3(a). The logic analyzer collects 2000 samples

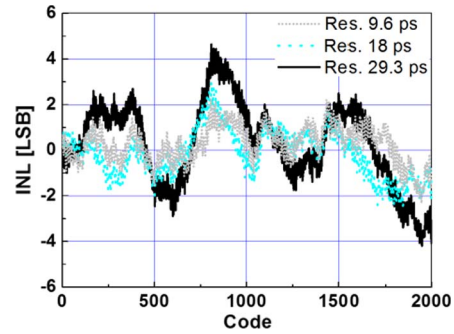
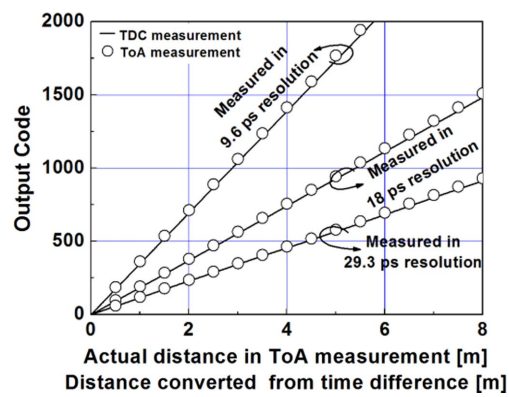


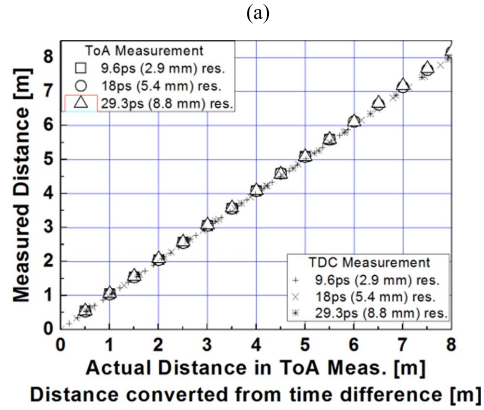
Fig. 12. INL measurements with 9.6 ps, 18 ps, and 29.3 ps resolutions.

TABLE II
PERFORMANCE SUMMARY OF TDC FOR ADAPTIVE IR-UWB RANGE FINDER

	Type / Value		
Architecture	Two-step TDC		
Bits	11		
Time resolution [ps]	9.6	18	29.3
Distance resolution [mm]	2.9	5.4	8.8
DNL_{max} [LSB]	0.65	0.76	0.94
INL_{max} [LSB]	± 2.3	± 3	± 4.6
Time error converted from INL_{max} [ps]	± 22	± 54	± 138
Distance error converted from INL_{max} [mm]	7	16	41
Converted maximum detectable range [m]	5.8	10.8	17.6
Power consumption [mW]	2.3		



(a)



(b)

Fig. 13. (a) The measured output code and (b) the measured distance depending on the actual/converted distance.

for a single-shot precision every 2-ps increase of the time difference between the two clocks at CLK_{rx} and Port_{ext}. The frequencies of both input clocks were 4 MHz. This setup is

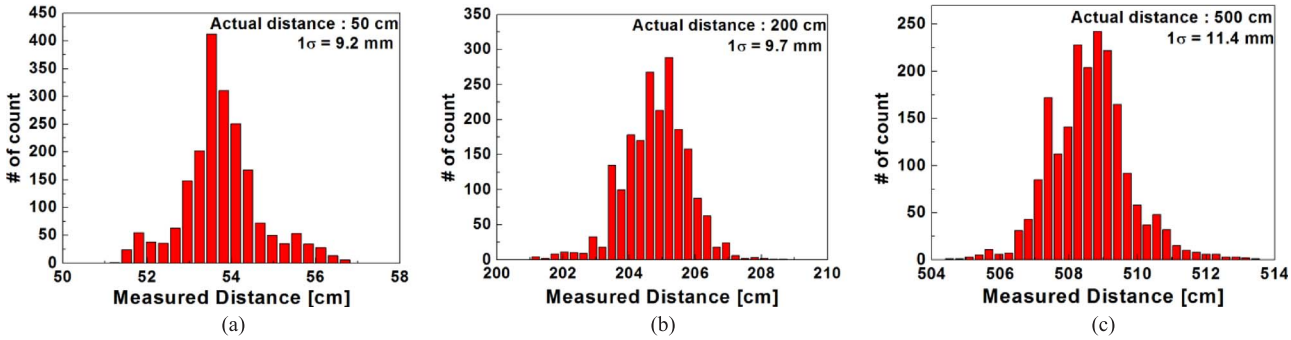


Fig. 14. Single-shot precisions in ToA measurement with 9.6 ps resolution at the distance of (a) 0.5 m, (b) 2 m, and (c) 5 m.

consistently applied for every measurement so that the adaptive IR-UWB range finder requires 500 μs (i.e., 2000 $\times$ 250 ns) for the decision, which is a sufficiently short time for the target applications.

Fig. 11 shows the TDC transfer curves measured with the time resolutions of 9.6 ps, 18 ps, and 29.3 ps. The x-axis and the y-axis indicate the time difference and the output code of the TDC. Note that the time difference between the two inputs of the TDC can be converted into the distance by multiplying the radio speed ($\approx 3 \times 10^8$ m/s). So the time resolutions of 9.6 ps, 18 ps, and 29.3 ps can be converted into the distance resolutions of 2.9 mm, 5.4 mm, and 8.8 mm. The distance converted from the time difference is expressed in brackets under the time difference on the x-axis. As can be seen, the maximum detectable time difference with a scalable resolution from 9.6 ps to 29.3 ps is adjustable from 20 ns to 60 ns, which corresponds from 6 m to 18 m in the distance domain, without any change in the dynamic range of 2000 steps. Fig. 12 demonstrates the measured INL. INL of a TDC can be estimated from (3). The maximum INLs of $\pm$2.3, $\pm$3, and $\pm$4.6 were measured with 9.6 ps, 18 ps, and 29.3 ps resolution respectively, which are expressed as the maximum time errors of $\pm$22 ps, $\pm$54 ps, and $\pm$138 ps. The shapes and the bumpy tendencies of the INL are broadly similar. However, the linearity visibly collapses in the case of 29.3 ps resolution, which implies that the main distortion occurs from the quantization process in the delay lines, not from the input network or the evaluation board [20]. This distortion is attributed to the foot transistor in the unit delay cell operating in weak inversion region, but can be prevented by modifying the circuit of the delay buffer. Logically, in the ranging system using a TDC, the INL of the TDC is directly connected to the error between the measured and actual distances [18]. Thus, the maximum time error of 22 ps corresponds to the distance error of 7 mm. Likewise, the maximum time errors of 54 ps and 138 ps correspond to the distance errors of 16 mm and 41 mm. Note that the maximum detectable ranges are measured 5.8 m with 9.6 ps resolution, 10.8 m with 18 ps resolution, and 17.6 m 29.3 ps resolution respectively. Fig. 14 shows the error distribution measured with 9.6 ps resolution when a 5-ns time interval was applied between two inputs of the TDC. In the Gaussian-shaped error distribution the average code is 518, and the standard deviation in time domain is 11 ps. The performance of the two-step TDC is summarized in Table II.

TABLE III
CALCULATED DISTANCE ERROR AND COMPARISON WITH MEASUREMENT RESULTS

τ_r [ps]	9.6	9.6	9.6	18	29.3
$\Delta D_{RX_thermal}$ [cm]	0.1	0.1	0.2	0.1	0.1
ΔD_{TDC_jitter} [cm]	0.42	0.43	0.5	0.4	0.38
ΔD_{TDC_INL} [cm]	0.58	0.58	0.58	1.08	1.77
ΔD_{walk} [cm]	3.4	4.6	8.1	4.6	4.6
ΔD_{Total} * [cm]	4.5	5.71	9.38	6.18	6.85
D_{real} [cm]	50	200	500	200	200
$D_{cal.} = D_{real} + \Delta D_{Total}$ [cm]	54.5	205.71	509.38	206.18	206.85
Average Measured Distance	54	205	509	206.4	207.8

* : $\Delta D_{Total} = \Delta D_{RX_thermal} + \Delta D_{TDC_jitter} + \Delta D_{TDC_INL} + \Delta D_{walk}$

C. ToA Measurement

The ToA measurement using the adaptive IR-UWB range finder has been tested up to actual 8 m with consideration for the indoor test environment. In the ToA measurement, we take average of 2000 pulses. The transfer curves from the ToA measurement are plotted and compared with those from the TDC-only measurement in Fig. 13. The measured distance can be calculated with TDC output code using following equation.

$$\text{Measured distance} = \text{TDC code} \times D_r \times c. \tag{12}$$

The empty round in Fig. 13(a) indicates the measured output code depending on the actual distance, and the lines indicate the output codes from Fig. 11. The distance between the two successive empty rounds is 50 cm, and the reference point "0 m" is actually 30 cm. In all cases of the measurements with the three resolutions, the measured output codes in the ToA measurement seem to approximately follow those in the TDC measurement, and are expressed in distance domain in Fig. 13(b).

To measure the error distribution in the ToA measurement, the single-shot precisions were performed at 50 cm, 200 cm, and 500 cm with 9.6 ps resolution. Fig. 14 demonstrates the Gaussian-shaped error distributions measured at the three positions. The standard deviations of the error distributions are 9.2 mm at 50 cm, 9.7 mm at 2 m, and 11.4 mm at 5 m respectively. As distance increases, the error standard deviation also increases. This can expected from (4). As distance increases, more stages in the TDC are needed to detect the signal which also increases $\Delta D_{\text{TDC_jitter}}$. Fig. 14 also shows that the average measured distances of each point (at distances of 50 cm, 200 cm, and 500 cm) were 54 cm, 205 cm, and

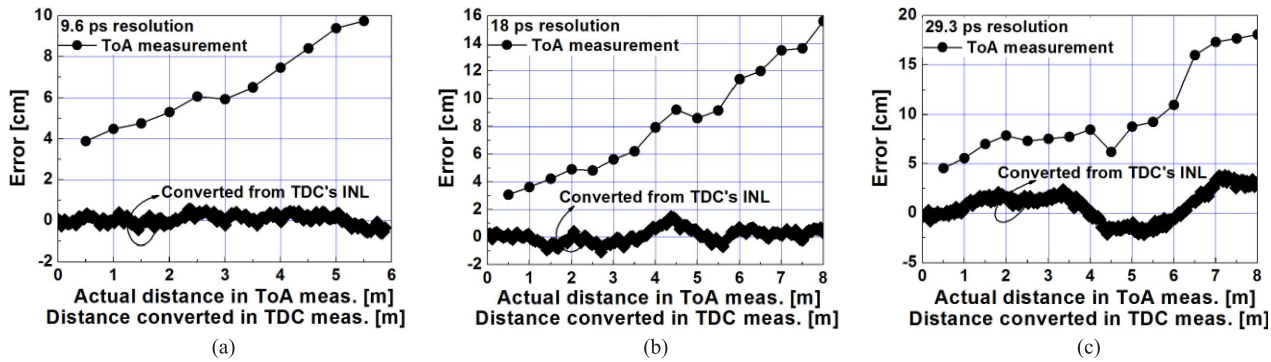


Fig. 15. Comparison of distance error in ToA measurement and distance error converted from the INL in TDC measurement.

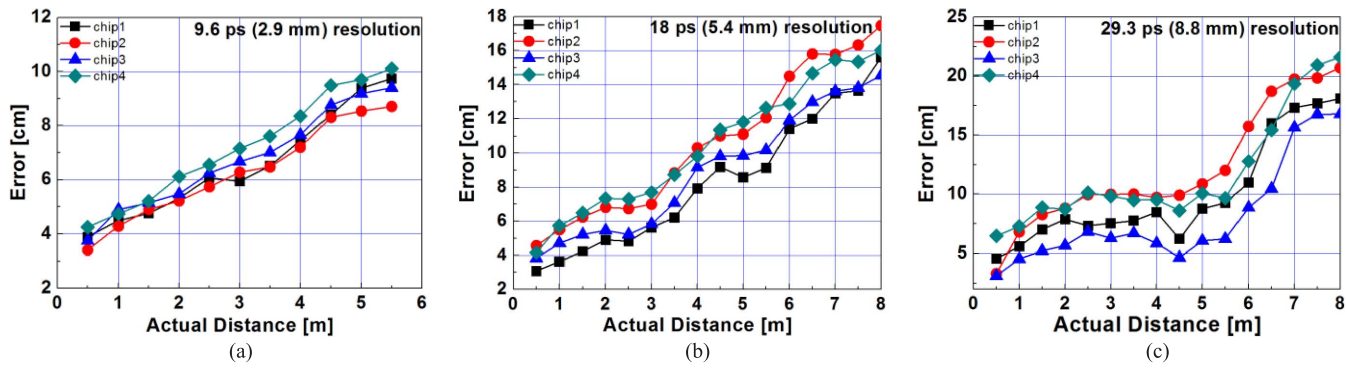


Fig. 16. Measured absolute distance errors of four range finders from one batch.

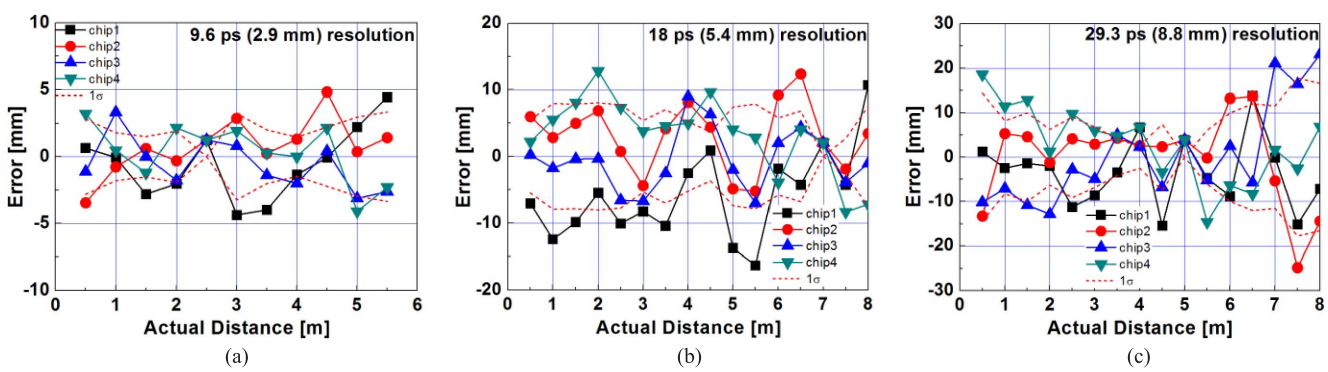


Fig. 17. Measured distance error and 1σ limits of four range finders after calibration.

509 cm respectively. The absolute error increased from 4 cm to 9 cm with the increase in distance.

Table III shows the individual contribution of each error source to the total error, and compares the measurement with the calculation results for various distance, and various values of τ_r. The contributions of $\Delta D_{\mathrm{TDC_INL}}$ and ΔD_{walk}, which are deterministic errors, are greater than those of $\Delta D_{\mathrm{RX_thermal}}$ and $\Delta D_{\mathrm{TDC_jitter}}$, which are random error. $\Delta D_{\mathrm{RX_thermal}}$ and $\Delta D_{\mathrm{TDC_jitter}}$ can be minimized by taking average of multiple pulses. ΔD_{walk} increases as distance increases as expected from (9) and (10); this explains the increase in absolute error over distance shown in Fig. 14. The ToA measurements were also performed for various τ_r, and the obtained errors are plotted along with those of the TDC measurements in Fig. 15. According to (2) and Table III, as τ_r increases, the contrbution of $\Delta D_{\mathrm{TDC_INL}}$ should increases; in fact, this increase in the TDC INL contribution with increasing τ_r can be clearly seen in Fig. 15.

Both the walk error and the TDC error can be affected by PVT variations. This result from the facts that the receiver gain varies with PVT, thus affecting the walk error, and that the TDC's delay line is composed of an inverter chain, is also affected by PVT variations. Therefore, to maintain a constant performance in the presence of PVT variations, proper compensation techniques should be applied [29].

D. Digital Calibration and Trimming for Inaccuracy

Digital calibration techniques are widely performed on devices that measure environmental quantities, with high accuracy compensating for the devices' main source of inaccuracy in sensing applications, such as those mentioned in [26]–[29]. To characterize the inaccuracy of the devices, four adaptive IR-UWB range finders from one batch were verified in the ToA measurement with the three resolutions. The batch calibration

is a well-known calibration technique in sensor applications [30]. The purpose of the batch calibration is to calibrate chip-to-chip variation. It takes average of few numbers of chips which is a batch and construct a master curve using the average. And the difference between master curve and the exact value is assumed an average error. The average error is used to calibrate other chips by subtracting the average error from distance measurement result of each chip. In our case, we construct a master curve with 4 chips. The individual calibration is conducted after the batch calibration for better accuracy. A single point is measured from each chip and the error between the single point measurement result and the average value from the batch is assumed to an offset. The offset is subtracted from the distance measurement result. In our case, the single point measurement is conducted at the distance of 4 m. The results are shown in Fig. 16. Each absolute error measured with the same resolution forms a curvature with a tendency similar to the others in the same group in terms of the value and the slope because all measurements are affected by both the ΔD_{walk} and the $\Delta D_{\mathrm{TDC_INL}}$. These errors can be digitally calibrated and compensated for by using a curvature correction method such as batch calibration [28].

For more precise accuracy of the sensor, individual trimming should be required after batch calibration [29]. Some error still remain among the tested range finders after batch calibration. This error are mostly attributed to the delay mismatch between range finders. The chip-to-chip variation results in a slope variation of the error over the distance. The mismatch can be trimmed to improve the error through the digital process of individual calibration. The slope parameter for each range finder was adjusted to cancel the error at a calibration distance where the error over the distance could be minimized.

The calibrated errors are shown in Fig. 17. After calibration process the resulting error (1σ) according to the resolutions are ± 3 mm, ± 8 mm, and ± 18 mm, respectively. These values are improved by 94%, 91%, and 83% compared to the absolute errors. Using this calibration method $\Delta D_{\mathrm{TDC_INL}}$ and ΔD_{walk}, chip-to-chip variation, are calibrated thus $\Delta D_{\mathrm{RX_thermal}}$ and $\Delta D_{\mathrm{TDC_jitter}}$ remain as source of error. The batch calibration and individual calibration can be included in mass production line which may increase test cost. Table IV summarizes performances of this works, with and without the calibration technique, and comparison with other studies. The major advantages of this work over other non-coherent UWB ranging systems such as [5] are a simple architecture for distance measurement and adaptive resolution for ranging. This work employs leading edge detection with a TDC as well as adaptive resolution by adjusting TDC resolution which allows the distance calculation with a simple and efficient hardware and achieves a fine resolution for short range at the same time covers long range with rough resolution.

TABLE IV
PERFORMANCE SUMMARY AND COMPARISON FOR TOA MEASUREMENT

	This work			[9]	[10]	[30]
Scheme	LED/ Scalable resolution			Correlator	QAC	EQ time Sampling
Technology	0.11-μm CMOS			0.18-μm CMOS	0.13-μm CMOS	0.13-μm CMOS
Resolution [ps]/[mm]	9.6 /2.9	18 /5.4	29.3 /8.8	.	.	.
$\mathrm{Range}_{\mathrm{max}}$ [m]	5.8	8 10.8 [a]	8 17.6 [a]	1	10	10
Absolute error [mm]	101	175	216	±25	±25 ±14 [c]	±15 [d]
calibrated error [mm]	-5~+5 ±3 [b]	-16~+12 ±8 [b]	-25~+23 ±18 [b]	.	.	.
Power [mW]	Tx: 1.5 Rx: 42			Tx: 0.7 Rx: 4	4.2	100
Area [mm^2]	Tx : 0.065 Rx: 0.27			Tx : 0.04 Rx: 0.38	0.22	3.27

LED: leading edge detection
EQ: equivalent
[a] : Maximum detectable range according to the TDC's dynamic range
[b] : 1σ
[c] : RMS error
[d] : Error compared with a commercial laser scope

VI. CONCLUSION

This paper introduces an adaptive CMOS IR-UWB range finder implemented with 0.11-μm CMOS technology for localization applications. The principle for the proposed ranging approach with scalable resolution was the practical process of the human recognition method. By virtue of scalable resolution, the proposed range finder based on the TDC can control its maximum detectable distance from 6 m up to 18 m without any change in dynamic range because of the scalable time resolution of the TDC. This resolution ranges from 9.6 ps (2.9 mm) to 29.3 ps (8.8 mm) and prevents restriction of the fixed maximum detectable distance. The most substantive and effective advantage of the adaptive CMOS IR-UWB range finder is its ability to use precise resolution to detect a short-range target, and rough resolution to detect a long-range target. In addition, the range finder makes the best use of delay cells in detecting the target, minimizing the number of redundant delay cells. The range finder has been verified over an actual distance to 8 m. Batch calibration to digitally compensate for the walk error and the INL of the TDC in the ToA measurements was followed by individual calibration to trim the slope for process mismatch in chip-to-chip variation. The final inaccuracies (1σ) according to the resolutions of 2.9, 5.4, and 8.8 mm were measured as ± 3 mm, ± 8 mm, and ± 18 mm, respectively. Moreover, as in [5] and [6], the 1D range detection in the ToA measurement with the proposed CMOS IR-UWB range finder can possibly be expanded to a 3D space location in a TDoA measurement with at least four receivers.

ACKNOWLEDGMENT

The authors thank Y. Kim, J. A. Lee, Prof. Y. Chae, and the School of Integrated Technology and Yonsei Institute of Convergence Technology (YICT) for their support.

REFERENCES

[1] A. Stelzer *et al.*, "A microwave position sensor with sub-millimeter accuracy," *IEEE Trans. Microw. Theory Tech.*, vol. 47, no. 12, pp. 2621–2624, Dec. 1999.
[2] R. Mosshammer *et al.*, "A 5.8 GHz local positioning and communication system," in *Proc. IEEE MTT-S Int. Microw. Symp. Dig.*, Honolulu, HI, USA, 2007, pp. 1237–1240.
[3] A. Hamidian *et al.*, "24 GHz CMOS transceiver with novel T/R switching concept for indoor localization," in *Proc. IEEE RFIC Symp.*, Jun. 2–4, 2013, pp. 293–296.

[4] C. Zhai *et al.*, "A 2.4-GHz ISM RF and UWB hybrid RFID real-time locating system for industrial enterprise Internet of Things," *Enterprise Inform. Syst.*, pp. 1–18, Mar. 2016.

[5] M. Mahfouz *et al.*, "Investigation of high accuracy indoor 3-D positioning using UWB technology," *IEEE Trans. Microw. Theory Tech.*, vol. 56, no. 6, pp. 1316–1330, Jun. 2008.

[6] C. Zhang *et al.*, "Real-time noncoherent UWB positioning radar with millimeter range accuracy: theory and experiment," *IEEE Trans. Microw. Theory Tech.*, vol. 58, no. 1, pp. 9–20, Jan. 2010.

[7] B. Waldmann *et al.*, "Method for high precision local positioning radar using an ultra wideband technique," in *Proc. IEEE MTT-S Int. Microw. Symp. Dig.*, Atlanta, GA, USA, 2008, pp. 117–120.

[8] Z. N. Low *et al.*, "Pulse detection algorithm for line-of-sight (LOS) UWB ranging applications," *IEEE Antennas Wireless Propag. Lett.*, vol. 4, pp. 63–67, 2005.

[9] T. Terada *et al.*, "A CMOS ultra-wideband impulse radio transceiver for 1-Mb/s data communications and ±2.5-cm range finding," *IEEE J. Solid-State Circuits*, vol. 41, no. 4, pp. 891–898, Apr. 2006.

[10] N. Van Helleputte *et al.*, "A reconfigurable, 130 nm CMOS 108 pJ/pulse, fully integrated IR-UWB receiver for communication and precise ranging," *IEEE J. Solid-State Circuits*, vol. 45, no. 1, pp. 69–83, Jan. 2010.

[11] M. K. Kang and T. W. Kim, "CMOS IR-UWB receiver for ±9.7-mm range finding in a multipath environment," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 59, no. 9, pp. 538–542, Sep. 2012.

[12] B. Merkl, "The Future of the Operating Room: Surgical Preplanning and Navigation Using High Accuracy Ultra-Wideband Positioning and Advanced Bone Measurement," Ph.D. dissertation, Dept. Biomed. Eng., University of Tennessee, Knoxville, TN, USA, 2008.

[13] M. Kuhn *et al.*, "High accuracy UWB localization in dense indoor environments," in *Proc. IEEE Int. Conf. UWB*, Hannover, Germany, 2008, vol. 2, pp. 129–132.

[14] M. Kuhn *et al.*, "A system-level design approach to UWB localization," in *Proc. IEEE MTT-S Int. Microw. Symp. Dig.*, Boston, MA, USA, 2009, pp. 1409–1412.

[15] M. Kuhn *et al.*, "Adaptive leading edge detection in UWB indoor localization," in *Proc. IEEE Radio Wireless Symp.*, New Orleans, LA, USA, 2010, pp. 268–271.

[16] K. K. Lee *et al.*, "A 5.2 pj/pulse impulse radio pulse generator in 90 nm cmos," in *Proc. IEEE ISCAS*, 2011, pp. 1299–1302.

[17] B. Qin *et al.*, "A single-chip 33 pJ/pulse 5th-derivative Gaussian based IR-UWB transmitter in 0.13 μm CMOS," in *Proc. IEEE Circuits Syst. Symp.*, 2009, pp. 401–404.

[18] Y. M. Kim and T. W. Kim, "An 11 b 7 ps resolution two-step time-to-digital converter with 3-D Vernier space," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 99, no. 2, p. 1, Feb. 2014.

[19] M. Lee and A. A. Abidi, "A 9 b, 1.25 ps resolution coarse-fine time-to-digital converter in 90 nm CMOS that amplifies a time residue," *IEEE J. Solid-State Circuits*, vol. 43, no. 4, pp. 769–777, Apr. 2008.

[20] L. Vercesi *et al.*, "Two-dimensions Vernier time-to-digital converter," *IEEE J. Solid-State Circuits*, vol. 45, no. 8, pp. 1504–1512, Aug. 2010.

[21] C. Weigandt, B. Kim, and P. R. Gray, "Analysis of timing jitter in CMOS ring oscillators," in *Proc. ISCAS*, Jun. 1994, vol. 4, pp. 27–30.

[22] L. Stoica *et al.*, "A low-complexity noncoherent IR-UWB transceiver architecture with TOA estimation," *IEEE Trans. Microw. Theory Techn.*, vol. 54, no. 4, pp. 1637–1645, Apr. 2006.

[23] S. Kurtti and J. Kostamovaara, "An integrated laser radar receiver channel utilizing a time-domain walk error compensation scheme," *IEEE Trans. Instrum. Meas.*, vol. 60, pp. 146–157, Jan. 2011.

[24] P. Dudek *et al.*, "A high-resolution CMOS time-to-digital converter utilizing a Vernier delay line," *IEEE J. Solid-State Circuits*, vol. 35, no. 2, pp. 240–247, Feb. 2000.

[25] J. Nissinen and J. Kostamovaara, "Wide dynamic range CMOS receivers for a pulsed time-of-flight laser range finder," in *Proc. IEEE Instrum. Meas. Sci. Technol. Conf.*, Como, Italy, May 2004, pp. 1224–1227.

[26] K. A. A. Makinwa, "Smart temperature sensors in standard CMOS," in *Proc. Eurosensors—Procedia Eng.*, Sep. 2010, pp. 930–939.

[27] K. Souri and K. A. A. Makinwa, "A 0.12 mm2 7.4 μW micropower temperature sensor with an inaccuracy of ±0.2 °C (3σ) from −30 °C to 125 °C," *IEEE J. Solid-State Circuits*, vol. 46, no. 7, pp. 1693–1700, Jul. 2011.

[28] K. Souri *et al.*, "A CMOS temperature sensor with an energy-efficient zoom ADC and an inaccuracy of ±0.25 °C (3σ) from −40 to 125 °C," in *Proc. IEEE ISSCC Dig.*, Feb. 2010, pp. 310–311.

[29] K. Souri *et al.*, "A CMOS temperature sensor with a voltage-calibrated inaccuracy of ±0.15 °C (3σ) from −55 °C to 12 °C," *IEEE J. Solid-State Circuits*, vol. 48, no. 1, pp. 292–301, Jan. 2013.

[30] P. Park *et al.*, "A centimeter resolution, 10 m range CMOS impulse radio radar for human motion monitoring," *IEEE J. Solid-State Circuits*, vol. 49, no. 5, pp. 1125–1134, May 2014.

[31] M. A. P. Pertijs and J. H. Huijsing, "Calibration technique," in *Precision Temperature Sensors in CMOS Technology*. Dordrecht, The Netherlands: Springer, 2006.

Tae Hwan Jin (S'13) was born in Seoul, South Korea, in 1983. He received the B.S. degree in electronic engineering from Kwangwoon University, Seoul, South Korea, in 2008, and a unified Ph.D. degree in electrical and electronic engineering from Yonsei University, Seoul, in 2015.

He is currently a Senior Engineer with System LSI, Samsung Electronics. His research interests are RF amplifiers, mixed-signal circuits, and IR-UWB systems.

Hong Gul Han (S'13) was born in Busan, South Korea, in 1984. He received the B.S and M.S. degrees in electronic engineering in 2010 and 2012, respectively, from Yonsei University, Seoul, South Korea, where he is currently pursuing the Ph.D. degree.

His research interests are in RF/analog circuits and systems for wireless application.

Tae Wook Kim (S'02–M'06–SM'10) was born in Seoul, South Korea, in 1974. He received the B.S. degree in electrical engineering from Yonsei University, Seoul, South Korea, in 2000, and the M.S. and Ph.D. degrees from the Korea Advanced Institute of Science and Technology (KAIST), Daejon, South Korea, in 2002 and 2005, respectively.

From July 2002 to December 2005, he was with Integrant Technology, Inc. From January 2006 to July 2007, he was with Qualcomm, Inc., Austin, TX, where he was involved with DVB-H and Media FLO chip design. Since September 2007, he has been with the School of Electrical and Electronic Engineering, Yonsei University, where he is currently an Associate Professor. His research interests are in microwave, RF, analog, and mixed-signal ICs and systems.

Dr. Kim is serving as a Technical Program Committee Member of IEEE ISSCC, IEEE APMC, IEEE A-SSCC, and as a Steering Committee Member of IEEE MWSCAS. He was an Organizing Committee Member of IEEE A-SSCC 2011 and IEEE MWSCAS 2011.